

Supplementary Information

Concentrated output sensitivity in connectome-scale biological models

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1 Measurement definitions and assay specifications

1.1 Sensitivity, spectral mass and numerical rank

An assay specifies a forward model, its parameter point, an input and an output vector. The sensitivity Jacobian is the derivative of that output in the stated parameter coordinates. We approximate it by finite differences and standardize output rows using reference scales. The positive semi-definite Gram matrix $H = J^\top W J$ measures squared output sensitivity. It has the Fisher-information interpretation only for a Gaussian observation model with known precision W ; most assays here use output-normalization weights.

The concentration counts d_{90} and d_{99} describe how many leading eigenvalues accumulate 90% and 99% of the trace. The participation ratio is $(\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$. These measures summarize the relative distribution of sensitivity. Numerical rank is evaluated separately from the complete spectrum at an explicit tolerance, and parameter recovery is assessed by fitting. Resolved weak directions can lie outside the selected concentration count. Silent individual Jacobian columns and insensitive linear combinations are also different objects.

For positive parameters, multiplicative steps $\theta_i(1 \pm \delta)$ and log steps $\theta_i e^{\pm \delta}$ both approach log-coordinate derivatives as $\delta \rightarrow 0$. At finite δ , they define distinct approximations. Signed biases require an affine scale, and zero-valued nonnegative parameters lie at a domain boundary. The visual-model analysis treats these cases explicitly (§4). Under a coordinate change $q = Tz$, the Gram matrix becomes $T^\top H_q T$; spectra therefore depend on the coordinate metric (1).

1.2 Primary assay specifications

Table 1: Assay specifications. Recordings in distinct rows need not use the same output metric or rollout. Their absolute eigenvalues are not directly comparable.

Assay	Parameters and input	Readout and normalization
Nematode connection weights	3076 weights; fixed sinusoidal sensory drive; relative step 0.5	80 motor-neuron voltages or 96 muscle outputs; 120 samples per output over 2000 ms; per-output temporal s.d., clipped at 10^{-3}
Displaced points	Same 300 connections at five parameter points; relative step 0.5	1500-ms voltage recordings; per-time across-neuron s.d.; fold widths 1.25 and 1.5
Conductance classes	Sixteen named mechanism gains; relative step 0.5	10,320 motor-voltage samples; per-neuron temporal s.d.
Closed-loop contexts	20 class gains or 14 neuron-class leak gains; step 0.25	Six world-frame path statistics per condition; reference-magnitude normalization
Fly walking controller	48 actuator gains; steps 0.05, 0.15, 0.25, 0.50	Ten summaries or 42 joint angles at 100 frames; baseline-magnitude normalization; five seeds in the step/readout comparison
Fly visual network	Bias, time-constant and synaptic-strength coordinates; explicit trained or initialized state	Last-five-frame neural activation mean during a 20-frame sinusoidal luminance movie, reduced to 64, 32, 16, 8 or 6 contiguous index bins
Cell models	Positive model parameters; log-space step 0.05 in the sensitivity assays	Settled voltage or concentration traces; features computed at full resolution; specified feature weights and fitting loss

For the closed-loop assay, the path vector contains displacement, path length, mean speed, speed s.d., r.m.s. heading change and the fraction of steps above the trajectory’s lower-quartile speed. The last component measures speed-quantile occupancy and is approximately fixed by construction. Path length and mean speed are proportional at a fixed number of steps. Components and repeated time samples need not be statistically independent. A combined assay stacks the recorded outputs and sums their Gram matrices using the specified condition weights.

2 Finite-difference and realization diagnostics

2.1 Nematode circuit

Two baseline simulations reproduce the full-assay voltage exactly. Four connections are selected by their full-assay column norms: strongest, median, lower quartile and weakest nonzero. Across relative steps 0.05, 0.1, 0.25 and 0.5, the step-0.5 derivatives differ from the step-0.05 estimates by 0.0061%, 0.196% and 8.27% for the first three columns. The weakest column is zero at 0.05 and gives the same quantized response at the three larger steps. Its derivative is unresolved. These 34 rollouts support the sampled strong and median responses. The complete 3076-column spectrum has not been evaluated at smaller steps. Across the tested steps, the sampled strong and median columns agree within 0.2%.

A separate connection control compares the standardized response to one weight perturbation at steps 0.05, 0.1, 0.25, 0.5 and 1.0. Response divided by step is approximately 2.84×10^{-4} throughout, with relative variation below 0.7%. Identical baseline simulations agree bit for bit. Repeatability and approximate proportionality hold for this sampled connection under its specified protocol.

2.2 Fly walking controller

At steps 0.05, 0.15, 0.25 and 0.50, response divided by step is 1.98, 1.39, 1.34 and 1.31. Joint-angle d_{90}/d_{99} values are 16/38, 12/31, 11/31 and 11/29; coarse-summary values are 1/1 throughout. The largest step has 10 divergent simulations among 1925 runs, and two gains lack a usable perturbation pair. The reported comparison uses step 0.25.

A fixed-parameter seed control has $d_{90} = 7$ and $d_{99} = 22$. These counts describe the concentration of seed variability; they provide no rank subtraction for the parameter-response counts 11/31. A separate summary-statistic assay gives 3/6 at step 0.05 instead of 1/1; the implementation difference remains unresolved. It does not enter the matched step-0.25 output comparison.

2.3 Oscillator and discontinuous locomotor assays

In the pyloric implementation, a pointwise trace response is approximately constant across a 250-fold perturbation range. Its finite-difference Jacobian consequently scales approximately as the inverse step. Phase shifts and finite recording windows can produce this response, so the tested pointwise differences do not resolve a stable local derivative. The tested phase-invariant summaries also fail response proportionality: the response-to-step ratio varies by 23.2 across a 25-fold step range. We use the accepted parameter population to describe parameter variation compatible with the selected rhythm (§6).

The modWorm posture response is discontinuous under the tested implementation and stimulus. A synaptic rise-time perturbation produces negligible change at 0.1% and 1%, then an output change of 0.141 at both 5% and 10%. Two of seven mechanism gains have changes near numerical precision at the finite step. Its sensitivity counts describe finite-step responses under these settings.

Repeated larvaworld baselines show substantial realization variability. The stored median perturbation signal-to-noise ratio is 1.21, and three tested parameters are below their own baseline-variation floor. Its single-realization spectrum is excluded from the primary concentration comparison.

Result files. Results: E42_fd_validation.json and E42_fd_validation_arrays.npz; E44_baai_fd_controls.json; AB4_flygym_delta.json, E36b_flygym_validstep.json; Seed-variation spectrum ($d_{90} = 7$, $d_{99} = 22$): E1_flygym_null.json; summary assays: E1_flygym_rank.json, flygym_hessian48.json; AB2b_stg_scaling.json, AB2d_stg_whitening.json, E33_stg_fd_convergence.json; modworm-REAL_b1.json, E7_E11_modworm.json, CP_coupling.json; LARVA_noise_floor.json.

3 Motor-output sensitivity and stimulation contexts

3.1 Connection weights and output layers

The fixed-input assay drives the fifteen sensory inputs with presynaptic voltage $u_i(t) = 2\sin[2\pi(0.5t + i/15)]$ in mV, with t in seconds. NEURON uses $dt = 5/3$ ms, $v_{\text{init}} = -65$ mV and $\text{secondorder} = 0$, over 2000 ms. The 80 motor outputs are sampled at 120 points each. Relative connection steps are ± 0.5 ; the matrix records the centered output difference divided by the perturbation interval in dimensionless relative coordinates. Output rows are divided by each baseline neuron’s temporal standard deviation, floored at 10^{-3} mV. The complete connection-weight assay gives 26/66, participation ratio 6.83 and no column below the recorded response threshold. At the floating-point Gram tolerance, 932 eigenvalues are numerically resolved. The tolerance measures arithmetic resolution; parameter recovery is evaluated by fitting.

An earlier assay with incompletely specified input and rollout gives 8/47 and participation ratio 2.55. Applying its float32 arithmetic to the fixed-input responses yields 26/66, so numerical precision does not explain the discrepancy. The equivalence of these two assays cannot be established. The primary spectrum uses the fully specified fixed-input assay.

The drive sets the synapse’s presynaptic voltage. Each external input connection has fixed dimensionless gain 10; these added input gains are outside the perturbed internal connection vector.

An individual column is classified as numerically inactive when

$$\|K_i\|_2 < 10^{-12} \max(\max_j \|K_j\|_2, 10^{-30}).$$

No step-0.5 column meets this arithmetic cutoff. The smallest column remains unresolved in the smaller-step control in §2.

The internal connection vector contains 1341 negative inhibitory-synapse weights, 651 positive synapse weights and 1084 positive gap-junction weights, with no zero weight. Multiplicative steps preserve each sign. Their relative coordinate locally changes log magnitude while preserving connection sign.

Chemical synapses contribute 69% of squared motor-voltage sensitivity from 1992 weights, compared with 31% from 1084 gap-junction weights. The mean squared sensitivity per chemical connection is 1.22 times the gap-junction value, while the median column norm is 6.62 times greater. The difference between these ratios shows that typical connection sensitivity and total class contribution describe different aspects of the distribution. These measurements use the fixed-input voltage assay. The overview trace displays baseline motor voltage after the initial transient, using the neuron with the largest temporal standard deviation over the displayed window.

The muscle readout maps motor voltages to preactivation Vw and muscle activation $\text{clip}((Vw + 80)/100, 0, 0.8)$. Both are derived from the same motor-voltage perturbation rollouts using the model’s retained readout weights. Each has $96 \times 120 = 11,520$ output components and its own per-muscle temporal standard deviation, floored at 10^{-3} in that observable’s units. Complete spectra give 3/9 for preactivation and 5/17 for clipped activation. The contrasts reflect both the output map and its reference normalization. The separate earlier muscle-readout count 3/11 has no verified clipped-activation interpretation.

Each displaced parameter vector multiplies every weight independently by a factor drawn uniformly in log space between the reciprocal and the stated fold width; replicate labels denote independent draws. The same 300-connection subset at the reference and four displaced parameter points has $d_{90} = 8\text{--}12$ and $d_{99} = 28\text{--}30$. Mean pairwise leading-subspace cosine is 0.89, with range 0.78–1.00; the matched random control is 0.162 ± 0.019 . The full-vector displacement changes every weight, but the sensitivity matrix is recomputed only on this common subset. This assay uses a different rollout and output normalization from the full-connectome assay.

3.2 Named conductance gains

Each conductance variable is assigned to its containing NEURON membrane mechanism. In particular, the same SLO conductance variable can occur in distinct EGL-19- or UNC-2-coupled mechanisms. Each gain acts on the conductance variable within its specified membrane mechanism. Uniform class gains change all compartments expressing the named mechanism together.

The sixteen-gain assay gives $d_{90} = 3$, $d_{99} = 6$ and participation ratio 2.43. Its largest column norms are IRK 123.5, NCA 117.7, EGL-36 86.0 and EGL-2 44.6; EGL-19 is 1.77 and UNC-2 0.17. Every gain responds. The complete eigenvalues range from 2.26×10^4 to 1.50×10^{-3} , above the stated floating-point tolerance of 2.26×10^{-8} . Eigenvector participation ranges from 2.88 for the leading combination to 1.02 for the weakest. A floating-point tolerance is distinct from a finite-difference convergence bound. These values characterize finite-step responses; the conductance-class spectrum has not been tested across perturbation steps.

Per-cell granularity changes the interpretation of uniform gains. A modWorm finite-step leak probe over 279 neurons gives $d_{90} = 1$ and $d_{99} = 2$, but its uniform direction carries only 0.295% of the trace. This is a finite-step finding under the discontinuous implementation described in §2. The class-gain ranking therefore cannot specify which cells carry the effect.

3.3 Combined stimulation conditions

The paired sensory-versus-motor perturbation comparison gives sensory output changes 0.872 and 0.122 with food close and far, and motor changes 0.109 and 0.357. These quantities belong to this paired experiment; its baseline displacements differ from the separately computed class-gain union experiment.

The closed-loop assays perturb 20 global gains or 14 neuron-class leak gains in log coordinates at step 0.25. Each of two food locations supplies six path statistics over 120 physics updates. Rows are normalized by $|y_i^{\text{baseline}}| + 10^{-9}$ separately for each condition. Stacking these standardized Jacobians gives K_{union} and $H_{\text{union}} = H_{\text{chemo}} + H_{\text{loco}}$. Both assays use the same closed-loop physics implementation. The path statistics of both reference trajectories reproduce the previously reported baselines exactly.

At relative singular-value thresholds 10^{-4} and 10^{-6} , each experiment has 4 and 4 single-condition directions and 8 combined directions. Floating-point cutoffs also resolve small residual singular values; this arithmetic resolution differs from a measured simulation-noise floor. At the

reported tolerance, the complete spectra resolve eight combined sensitivity directions. The 20-gain counts are 2/4, 1/2 and 2/4; the 14-class counts are 2/4, 1/2 and 1/4, respectively. Combined eigenvalues beyond rank four carry 0.73% and 0.30% of their total squared sensitivity. These tail fractions use the entire spectrum and do not identify a fixed set of newly added parameter directions.

In the combined-condition eigenvalue panel, values below 10^{-10} of each leading eigenvalue are omitted on the logarithmic axis without substituting a plotting floor. All twenty eigenvalues enter the concentration-count denominators. Numerical ranks use singular-value tolerances, separately from the plotting cutoff.

For four FlyGym descending commands, subset-averaged combined d_{90} values are 12.25, 12.67, 12.50 and 13.00 as the number of commands increases from one to four; corresponding d_{99} values are 30.75, 31.67, 32.00 and 32.00. Error bars give the standard deviation across command subsets. These command variations share a controller and action family.

Result files. Implementation: `create_environment / physics_step`. Results: `E42_full_spectrum_rerun.json`, `E42_provenance.json`, `E42_motor_voltage_matrices.npz`, and previous `E42_baaiworm_connectome.json`; `E42_full_spectrum_rerun.json`, `E42_muscle_preactivation_matrices.npz`, `E42_muscle_activation_matrices.npz`; previous `E60_baai_muscle_full.json`; `E58_baai_multipoint.json`, `null_subspace_overlap.json`; `E45b_baai_named_channels.json`, `AB7_baai_participation.json`; `E37_granularity_modworm.json`; `BAAI_chemo.json`; `B6_full_union_rerun/result.json`, `B6b_full_union_rerun/result.json`, their `full_matrices.npz` and rollout archives; `Historical B6_union_crossbehaviour.json` and `B6b_cellclass_union.json` contain only six eigenvalues per joint spectrum; `E36b_flygym_vali_dstep.json`.

4 Visual-network state and parameter coordinates

The analysis uses flyvis 1.1.2 and one official pretrained flow network. Model identity and library versions are specified with the result files. The matching initialized control uses the same connectome, stimulus, parameter-index ordering and fixed bias scale. Each simulation first computes a one-second grey steady state, then a twenty-frame sinusoidal luminance movie at 0.02 s per frame. Neural activations are averaged over the last five frames and reduced to contiguous neuron-index bins. The luminance input is $0.5 + 0.5 \sin[2\pi(i/40 - 0.1f)]$, with flattened input index i and frame index f ; it is not assigned a verified two-dimensional spatial geometry. Each output bin groups a contiguous range of neuron indices.

All 734 parameters are perturbed. The primary subset omits the 31 synaptic strengths that are zero in the trained state, leaving 703 shared interior coordinates. Positive parameters satisfy $\theta = \theta_0 e^q$; biases satisfy $\theta = \theta_0 + q$ with fixed unit scale. At a zero strength, the separate boundary analysis uses $\theta = sq$, $q \geq 0$, with s equal to the corresponding initialized strength and a second-order right derivative. Its Gram matrix measures sensitivity along the feasible, one-sided boundary directions.

For each bin count, the state comparison uses the trained reference scales $s_i = |y_i^{\text{trained}}| + 10^{-6}$ for both states. At 64 bins, the matched initialized and trained counts are 15/34 and 9/22. At 32, 16, 8 and 6 bins, they are respectively 3/11, 2/8, 2/6, 1/5 for initialization and 2/7, 2/5, 1/5, 1/4 for the trained network. Normalizing each state by its own baseline instead gives 43/58 for initialization and 9/22 for the trained state. The primary state comparison holds the trained-reference weights fixed; state-specific normalization defines a separate comparison.

At 64 bins, all 64 singular directions exceed the tolerance $\epsilon_{\text{mach}} \max(n, p)\sigma_1$. The concentration counts 9/22 summarize how squared sensitivity is distributed within this resolved span. Including the boundary directions leaves the trained 64-bin counts unchanged. The result files include complete spectra, raw and standardized Jacobians, parameter definitions and readout boundaries for each step and bin count.

The trained concentration counts remain 9/22 at parameter steps 0.003, 0.01 and 0.03, with aggregate Jacobian differences below 1%. With the physical movie and output scales fixed, integration steps 0.02 s, 0.01 and 0.005 s give concentration counts 8/22 and 8/21 at the two finer steps. The trained Jacobian differs from its step-0.01 estimate by 0.58% at step 0.003 and 0.81% at 0.03. At the smaller step, 55 full-matrix columns differ by more than 10%, including numerical-zero and weak columns; they account for 0.0026% of primary squared sensitivity. These columns contribute little to total squared sensitivity, although their individual derivatives remain unresolved. Repeat baseline differences are at most 6.7×10^{-16} . The neuron dynamics use $\max(\tau, dt)$; 51 trained time constants lie below the 0.02-s step, so some weak derivatives reflect this implementation floor. Activation transitions can also affect finite-step derivatives. Integration-step convergence is assessed separately below.

A separate protocol gives 14/32 using 330 raw coordinates in an initialized network.

Posthoc integration controls hold the same sinusoidal luminance movie for 0.4 s, the initial grey input for 1 s, each original frame for 0.02 s, and the final averaging window for 0.1 s. Using the same 703 coordinates and the output scales from the native 0.02-s baseline, steps 0.02, 0.01 and 0.005 s give 9/22, 8/22 and 8/21. Jacobians at the finer steps differ by 8.90% and 11.11% from the native estimate, and by 2.56% from each other. These controls also release time constants held at the native implementation floor, so they change the effective implemented dynamics as well as integration resolution. The trained model uses 0.02 s. Renormalizing each finer-step output by its own baseline instead gives 2/3 and 1/2, because some bin means shrink substantially; these changed-weight results are not used for the fixed-metric integration comparison.

Result files. Results: `flyvis_trained_dimensionless.json`, `flyvis_initialized_dimensionless.json`, `flyvis_state_coordinate_comparison.json`, `flyvis_verification.json`, `E4_flyvis_readout.json`; `flyvis_timestep_control.json` and its complete control matrices.

5 Known-truth fitting and recovery

5.1 Model panel and fitting protocol

The panel contains five neuronal models and six biochemical oscillators: Hodgkin–Huxley, A-current Hodgkin–Huxley, Wang–Buzsáki, Morris–Lecar and FitzHugh–Nagumo; MAPK, repressilator, Goodwin, Selkov, Brusselator and Goldbeter (2–12). Equations, fixed constants, initial conditions and output indices are specified in the accompanying implementations. Generating traces are produced by these implementations; the recovery experiment tests their parameter inference under the stated assays.

Generating parameters are the specified model defaults. The random start is $\theta_i^* e^{\eta_i}$, with η_i independently uniform in $[-\ln f, \ln f]$ for $f = 1.5, 3$ and 10 . This distribution uses the truth as its center. Settled traces supply amplitude quantiles, Fourier magnitudes and autocorrelation features computed before reduction. Most models discard 40% of the trace; the A-current neuron discards 50% to avoid its silent interval. CMA-ES uses population 16, at most 400 iterations, and the specified seed-dependent initialization. Loss below 0.02 defines fit success.

The reported parameter error is $\|\hat{\theta} - \theta^*\|_2 / \|\theta^*\|_2$, computed in the units of the implemented parameter vector. This metric is not invariant to rescaling individual coordinates with different units. We compare recovery within each model using this error and the per-parameter ratios. Parameter medians in the panel condition on successful output fits, and fit fractions are shown separately. In the cross-model parameter-recovery plot, marker area is $15 + 45r$ points squared, where r is the fraction of successful fits. Errors are plotted at their measured values without a plotting floor.

5.2 Recovery outcomes and the MAPK trace example

The Hodgkin–Huxley fit succeeds on 72/72 starts with median normalized error 1.85×10^{-5} . The A-current and Wang–Buzsáki fits succeed on 15/72 and 39/72, with errors below 3×10^{-5} among successful fits. Morris–Lecar succeeds on 17/24 and FitzHugh–Nagumo on 15/24. MAPK succeeds on 5/24 with median normalized parameter error 0.40. These results measure the performance of the specified assay, starts and optimizer.

The displayed MAPK trace has parameter error 0.4266 and feature loss 0.02783. It is outside the 0.02 success threshold and is shown as an approximate fit. The conditional MAPK summary instead uses the five successful fits. The generating trace follows the implemented equations over the 6000-s simulation window; the time axis uses the corresponding sample count.

5.3 Parameter-count and observation grid

The Hodgkin–Huxley grid frees three conductances; five parameters after adding leak reversal magnitude and capacitance; or eight after adding gating-rate scalings. Its output choices are one spike-rate statistic, seven current–frequency and spike-shape features, and voltage traces of 200 or 1000 samples. Spike counts are discrete, so their finite-step sensitivity requires the same distinction from an infinitesimal derivative as other nonsmooth readouts.

All twelve grid cells have measured $d_{90} = 1$. Median parameter errors for the seven-feature assay are 3.8%, 7.3% and 38% as the free parameter count increases; median output losses are below the fit threshold. Trace-assay medians fail the output threshold, with parameter errors up to 56%. Daggers mark grid cells whose median output loss meets or exceeds the fit criterion.

5.4 Coupled-cell size control

Coupling identical Hodgkin–Huxley cells with three free conductances per cell holds the model family and observable budget approximately fixed per parameter. From 6 to 192 parameters, d_{90}/p changes from 0.33 to 0.30 at the lower budget and from 0.50 to 0.31 at the higher budget (Fig. 1). This bounds a simple parameter-count explanation within this constructed family. The ratio normalizes the concentration count by parameter number.

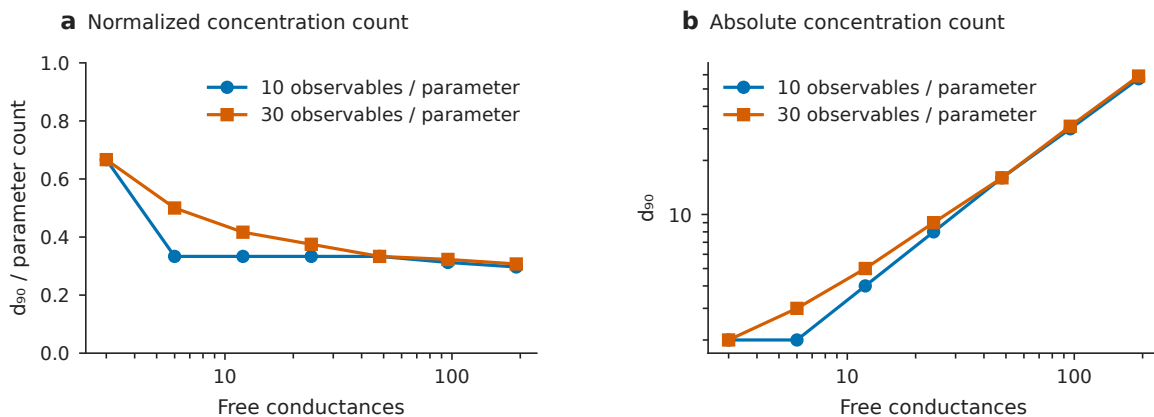


Figure 1: **Sensitivity concentration in coupled identical neurons.** **a**, d_{90} divided by the number of free conductances. **b**, Absolute d_{90} for the same runs. Blue circles denote the target budget of ten observables per parameter, and vermillion squares thirty. Lines connect simulated network sizes, including the single-cell, three-conductance reference. Integer output counts approximate the target observable budgets.

Result files. Results: `cell_panel13_result.json`, `cell_panel13b_result.json`, `cell_stationary.json`; `mapk_showcase.json`, `mapk_trace_metadata.json`; `cell_ladder_result.json`, `cell_diag_result.json`; `cell_scale_result.json`.

6 Biological context and exploratory animal comparisons

6.1 Model populations and empirical variation

The Prinz–Marder accepted model population provides a direct example of parameter variation compatible with a selected rhythm (13). Across 2365 accepted sets, the 31-coordinate cloud has PCA participation ratio 23.2, with 23 and 30 components reaching 90% and 99% of its variance. This PCA summarizes covariance among the accepted parameter vectors.

Across 1914 Allen cortical neurons, eleven electrophysiological properties have PCA participation ratio 4.78. Coefficients of variation for non-voltage properties span 0.34–1.85; voltage coefficients depend on the arbitrary zero and are excluded from that comparison. Across forty nematode channel and synaptic genes, ortholog sequence identity spans 77.9–99.9%, with `unc-9` highest. Across 61 isogenic N2 worms and 683 Tierpsy features, participation ratio is 9.75; centered covariance rank is at most 60. These quantities concern electrophysiological heterogeneity, sequence conservation and behavioral variation, respectively. They characterize distinct biological levels of description (14–16).

6.2 Two fly connectomes

An exploratory comparison based on the original visual-model ranking tests whether connections among the listed strongly sensitive types are more consistent across a female FlyWire and a male CNS specimen than connections involving the listed weakly sensitive types. Among connections present in both specimens, the two groups contain 34 and 29 type pairs. Weight discrepancy has

Mann–Whitney two-sided $p = 0.51$ in raw counts and $p = 0.96$ after fractional-input normalization. The comparison does not support the proposed ordering. Coverage, synapse-detection depth, sex and dependence among type pairs limit the interpretation. This test uses the earlier initialized-model ranking, separately from the trained-state analysis in §4.

6.3 Nematode signal propagation

An exploratory comparison with the signal-propagation atlas (17) uses rank-based neuron-class scores constructed from the closed-loop sensitivity lists. The primary sensory/interneuron subset comprises eight classes; the measured associations with response amplitude and significant-partner counts are not significant. For mean significant-response amplitude, Spearman $\rho = 0.203$ with exact two-sided $p = 0.641$. Head-focused coverage leaves most ventral-cord motor classes poorly tested.

Result files. Results: `E1_pyloric_manifold.json`; `E2_allen_variability.json`, `E4_channel_conservation.json`, `E5_n2_manifold.json`; `two_connectome_consistency.json`; `leifer_class_test.json`.

7 Additional analyses and reproduction

7.1 Postural modes and finite-step controls

Real nematode posture has participation ratios 3.79 and 4.14 under two skeleton resolutions; four leading modes account for 96% and 94% of variance (18). Model–data leading-subspace cosines are approximately 0.77 for modWorm and 0.68–0.69 for BAAIWorm. These comparisons describe posture variation in output space. The posture displacement ratio for stiff versus weak mechanism perturbations is 1.17, a modest descriptive contrast.

Finite-step repertoire probes in modWorm and a rodent controller have different subset curves under their respective parameter and output definitions.

7.2 Fit dispersion and posterior calibration

Multistart solution clouds measure dispersion among parameter fits at a specified optimization budget and output tolerance. A unique-optimum control also produces a dispersed cloud before optimization finishes, so the PCA dimension of these clouds does not establish an exact equivalence manifold.

Simulation-based calibration tests the calibration of an inferred posterior (19, 20). A prior-like posterior on an uninformative parameter can pass this test, so pass counts do not identify sensitive directions. The modWorm check evaluates posterior calibration under the specified simulation and inference procedure.

7.3 Reproduction scope

The accompanying data include the complete spectra and Jacobians available for each assay, its input and model state, and parameter-coordinate definitions. Figure scripts use these result files directly. Concentration counts can be recomputed from the matrices or complete spectra; forward simulations require the specified simulator installations. Section-level source lists identify the corresponding result files.

Result files. Results: EW_eigenworm.json, EW_baai_local.json, NCS_p1p5.json; SAT_saturation.json, MB_behaviour_specificity.json, RODENT_manifold.json; modworm_REAL_manifold.json, NCS_p1p5.json; modworm_REAL_sbc.json, D3_modworm_stats.json.

8 Prospective stimulus comparison

8.1 Synthetic observations and fitting

We generated 12 independent parameter vectors for the point-neuron circuit by drawing each of its 16 log conductance gains uniformly from $[-0.2, 0.2]$ around the published reference. Each voltage trajectory records 80 motor cells at 129 times over 1500 ms, with integration step 5/3 ms. Observations add independent Gaussian noise with standard deviation 1 mV. Each strategy receives one baseline trajectory and one additional trajectory. Within a case, strategies share the baseline observation and the same additional noise array; cases use independent noise arrays. The primary endpoint is voltage root-mean-square error (RMSE) against the noiseless generating response to one excluded test stimulus. The secondary endpoint is RMSE in natural-log conductance gains.

Fits minimize half the summed squared voltage residuals divided by the noise variance, plus a Gaussian log-gain penalty with standard deviation 0.35. Each log gain is bounded by $\pm \ln 2$. The baseline uses the public reference and an independently perturbed reference as its two starts. All additional-data fits share the best baseline fit and the second baseline fit as their two starts; these can coincide. The trust-region solver uses central differences at log step 0.001, initial radius 0.05, maximum radius 0.2, and at most 50 iterations per start. A proposed step must reduce the actual objective and achieve an actual-to-predicted improvement ratio above 0.1. It stops after a small improvement at training RMSE at most 1.02 mV, a step smaller than 0.001 at RMSE at most 1.05 mV, no accepted step, or the iteration cap. Training RMSE above 1.05 mV marks a fit-quality flag, and flagged outcomes are included in all endpoint summaries. Simulation costs can differ despite equal measurement budgets and iteration caps.

8.2 Candidate stimuli and selection

The baseline steps external presynaptic input to -5 mV in AWAL, AWAR, AWCL and AWCR from 250 to 1000 ms. The candidate library contains this stimulus and steps to -5 or $+5$ mV in each of five groups: the baseline group; ASKL, ASKR, PHAL and PHAR; ALNL and ALNR; PLML; and URYDL, URYDR, URYVL and URYVR. Two further candidates ramp to $+5$ mV in the baseline or ASKL/ASKR/PHAL/PHAR group over the same interval. Inactive input is -65 mV. Non-repeat strategies select from the 11 candidates other than baseline. The excluded test stimulus ramps URY inputs to $+2$ mV from 200 to 950 ms, then steps PLML to -2 mV from 1050 to 1250 ms. Neither this response nor the generating parameters enters selection.

For each baseline fit, we construct 16 compatible model vectors, including the best fit. Gaussian proposals use the inverse of the regularized local sensitivity precision, with fixed proposal scales 1, 0.5 and 0.25. Proposals must satisfy the parameter bounds and have nonlinear penalized objective within 13.148 of the best fit. These proposals form an approximate baseline-compatible ensemble; they are not exact posterior samples. For a candidate, let C be the sample covariance of its predicted voltage vector across these models. The ensemble-selection strategy maximizes $\frac{1}{2} \log \det(I + C/\sigma^2)$, evaluated through the equivalent model-space determinant, where $\sigma = 1$ mV. This is a Gaussian ensemble information proxy. The largest-response control maximizes root-mean-square voltage change from the corresponding unstimulated responses, averaging squared changes across models,

motor cells and time. The other controls repeat the baseline or choose a candidate uniformly at random.

8.3 Results and uncertainty

All 12 cases completed baseline fitting and ensemble construction, giving 48 strategy outcomes and 24236 retained forward responses. Table 2 includes all outcomes. One ensemble-selected fit ended at the iteration cap with training RMSE 2.622 mV after 50 iterations. Its excluded-stimulus RMSE was 0.986 mV. Figure 2 displays all cases. Uncertainty intervals use 10000 paired bootstrap resamples of the independent parameter cases, preserving the shared noise within each case. The independent parameter cases are the statistical replicates. The 95% intervals describe each comparison separately.

Table 2: Prospective strategy outcomes. Parameter errors are in log gains; all training-quality flags are retained.

Strategy	Mean voltage RMSE (mV)	Mean parameter RMSE	Flags
Ensemble selection	0.11515	0.08206	1/12
Repeat	0.05323	0.07435	0/12
Random	0.04891	0.07717	0/12
Largest response	0.03908	0.07393	0/12

The mean paired voltage-error differences for ensemble selection minus repetition, random selection and largest response were 0.06192, 0.06624, 0.07607 mV, with 95% intervals $[-0.04079, 0.23164]$, $[-0.03207, 0.23424]$, $[-0.00536, 0.23526]$ mV, respectively. All intervals cross zero, and average errors are higher for ensemble selection. The comparison provides no demonstrated improvement or equivalence among these strategies for synthetic Gaussian observations near the reference parameters, the excluded waveform and the specified approximate ensemble and solver.

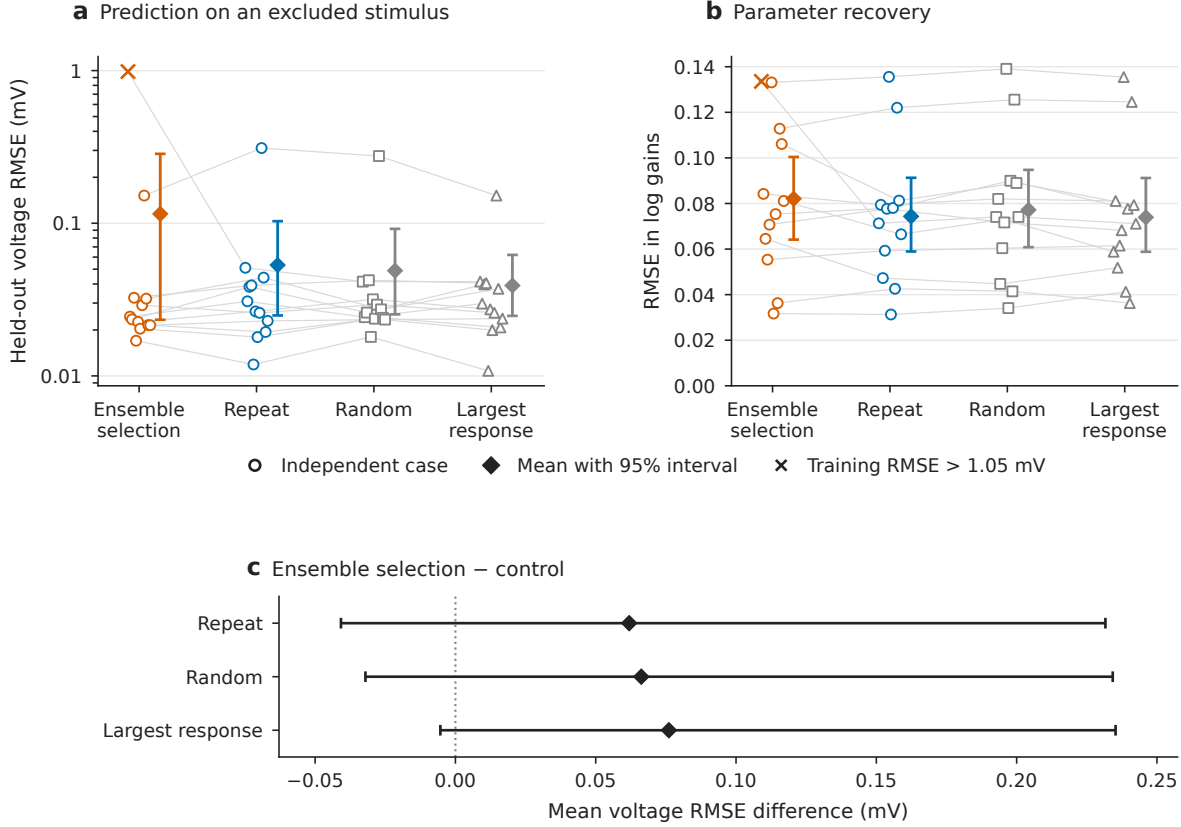


Figure 2: **Prospective stimulus comparison across independent simulated cases.** **a**, Voltage RMSE against the noiseless test response, displayed on a logarithmic axis. **b**, RMSE in natural-log conductance gains. Open markers show 12 independent parameter cases. Light gray lines link the same case; horizontal offsets separate overlapping points. Vermillion circles denote ensemble selection, blue circles baseline repetition, gray squares random selection and gray triangles the largest ensemble root-mean-square voltage change from unstimulated responses. Crosses replace markers for training RMSE above 1.05 mV and remain in every summary. In panels a and b, filled diamonds, offset right, show arithmetic means; whiskers show pointwise descriptive 95% percentile intervals from 10000 bootstrap resamples of independent cases. These are intervals for mean endpoints, rather than paired strategy differences. Panel a summaries are computed in mV before logarithmic display. Each case has one baseline and one additional noise realization shared across strategies. All 48 outcomes are shown. **c**, Paired differences in mean voltage RMSE between ensemble selection and each control. Black diamonds and horizontal whiskers show the means and paired bootstrap intervals specified above; the dotted gray line marks zero. Negative differences favor ensemble selection.

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